

Metrics Match Consequences

Project: Responsible Plant Classifier

1. Metrics Match Consequences

Teach the learning cycle

Primary target: Apply a declared score threshold including equality

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

Look for: Select a model using a declared validation rule

Reference example, not the only acceptable wording: Teacher explanation: Choose A: validation MAE 3 ties C, and A wins the simplicity tie. B has validation MAE 5 despite training MAE 0.

Train a Regression Model

Without opening the old lesson, explain one key idea from Train a Regression Model. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a line from a declared training candidate set

Reference example, not the only acceptable wording: Teacher explanation: A 5 B 3 C 0; choose m 2 b 1.

Regression and Residuals

Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish numeric prediction from classification

Reference example, not the only acceptable wording: Teacher explanation: Residuals -1, 1, 0 cm; the first record is overpredicted.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A reviewer can inspect at most 10 alerts daily. Model A flags 20 cases with higher recall; B flags 8 with lower recall. What evidence is needed before choosing a threshold?

Concept to explain

Metric choice depends on intended use, the positive class, who is affected, and the consequences of each error.

Term · Meaning

Term: consequence

Meaning: real effect of a prediction error

Term: threshold

Meaning: score boundary for a class decision

Term: cost matrix

Meaning: relative impact assigned to error types

Term: metric suite

Meaning: multiple measures covering distinct concerns

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on thresholds and consequences. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

All reference positives, TP+FN.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Optional Python 3 editor; no external packages or AI accounts required

End goal for the pictorial case

By the end, I can explain metrics match consequences using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Select metrics for screening, recommendation, and forecasting tasks.

MODEL IDEA Metric choice depends on intended use, the positive class, who is affected, and the consequences of each error.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Screening task treats missed positives as costly, so emphasize recall and human follow-up. Recommendation task may track precision among shown items plus user benefit.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Screening task treats missed positives as costly, so emphasize recall and human follow-up.

STEP 2 Recommendation task may track precision among shown items plus user benefit.

STEP 3 Height forecast uses MAE in centimeters and residual slices.

STEP 4 No metric replaces a written consequence analysis and subgroup review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

Connect metrics to consequences.

Exact explanation to model

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using consequences, review capacity and validation evidence.

Explain the added worked example

A classifier score is not automatically a probability. Declare blocked positive and predict blocked when score is at least the threshold. Changing this boundary changes decisions without necessarily retraining the model. Select it using development evidence and a task-relevant rule, not final-test answers.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines thresholds and consequences.

Six records have score/reference pairs A .9/blocked, B .7/clear, C .6/blocked, D .4/blocked, E .3/clear, F .1/clear. At threshold .5, TP=2, FP=1, FN=1, TN=2. At .3, TP=3, FP=2, FN=0, TN=1. Equality is positive, so E

is flagged at .3.

Trace the calculation on paper before running this optional Python check. Use the stated fictional inputs and compare intermediate values. The code checks this example; your independent workbook case uses different evidence.

Key terms

Term · Meaning

Term: Score

Meaning: A model output used to rank or decide

Term: Threshold

Meaning: A boundary that converts a score to a label

Term: False alarm

Meaning: A positive prediction for a negative reference

Term: Cost rule

Meaning: A declared way to value different mistakes

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thresholds and consequences at a glance

Part · Evidence or action · Interpretation

Part: Threshold .5

Evidence or action: One false alarm one miss

Interpretation: Four of six correct

Part: Threshold .3

Evidence or action: Two false alarms no misses

Interpretation: Four of six correct

Part: Decision rule

Evidence or action: Include equality

Interpretation: Record before evaluation

Lowering the threshold here finds D but adds E as a false alarm. The two accuracies are equal while consequences differ. Precision is $\frac{2}{3}$ at .5 and $\frac{3}{5}$ at .3; recall rises from $\frac{2}{3}$ to 1. This behavior follows these fixed scores and records, not a promise of future performance.

Give TP, FP, FN, TN at both thresholds. Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Metrics Match Consequences

Translate a threshold choice into missed-leaf and false-alarm costs.

The facts we will use

Scores/reference: 0.9 spotted, 0.6 healthy, 0.4 spotted, 0.1 healthy. Predict spotted at $\text{score} \geq \text{threshold}$.

Miss cost 5; false-alarm cost 1.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/25

Metrics Match Consequences

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Threshold 0.7

TP 1, FN 1, FP 0, TN 2: cost 5.



2. Threshold 0.4

TP 2, FN 0, FP 1, TN 1: cost 1.



3. Choose for stated costs

Threshold 0.4 has smaller validation cost on these four cases.

CHANGE ONE THING

Miss and false-alarm costs swap to 1 and 5.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Threshold0.7

What happens and why: TP 1, FN 1, FP 0, TN 2: cost5.

Step: 2. Threshold0.4

What happens and why: TP 2, FN 0, FP 1, TN 1: cost1.

Step: 3. Choose for stated costs

What happens and why: Threshold0.4 has smaller validation cost on these four cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Threshold0.4 has smaller validation cost on these four cases. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Toy costs illustrate decision design; real use needs agreed consequences, review capacity and fresh evaluation.

Change one thing in the pictured case

Change: Miss and false-alarm costs swap to1 and 5.

Prediction: Threshold0.7 costs1 while0.4 costs5; preferred threshold switches.

Reason: Metric choice encodes the declared consequences, not a universal best threshold.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: Spotting alerts can miss disease signs or create unnecessary healthy-leaf reviews.

Write the question to explain: Connect metrics to consequences.

Reveal after predictions: Recall measures finding positives; precision measures alert correctness. Set tradeoffs using consequences, review capacity and validation evidence.

Keep for an independent attempt: Missing a diseased plant is considered more costly than a false alarm, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: At .5: 2, 1, 1, 2. At .3: 3, 2, 0, 1. Exit response: Selection uses the test answers, turning the test into development data and weakening its independent evaluation role.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: Which denominator defines recall? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify score in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: apply the boundary; count the consequences. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between score and threshold. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Calculate costs with FP cost 1 and FN cost 4.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Apply a declared score threshold including equality; Compare false alarms and misses at two thresholds. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Give TP, FP, FN, TN at both thresholds.

Teacher check: Teacher explanation: At .5: 2, 1, 1, 2. At .3: 3, 2, 0, 1.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Thresholds and consequences. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which labels determine the false-alarm count at threshold 0.4? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

The healthy 0.6 case is wrongly flagged; healthy 0.1 stays clear. There is exactly one false alarm. Changed-case check: Threshold 0.7 costs 1 while 0.4 costs 5; preferred threshold switches. Metric choice encodes the declared consequences, not a universal best threshold.

Changed-case reasoning: Threshold 0.7 costs 1 while 0.4 costs 5; preferred threshold switches. Metric choice encodes the declared consequences, not a universal best threshold.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Metric Council: map stakeholders and error costs, select a metric suite, and defend what each metric cannot show.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR The same highest-accuracy metric should choose every model.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

A stronger claim makes a stronger project: Selection uses the test answers, turning the test into development data and weakening its independent evaluation role.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: At .5 TP 1 FP 1 FN 1 TN 1 cost 5. At .2 TP 2 FP 1 FN 0 TN 1 cost 1. Equality is positive at both boundaries.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Recall measures finding positives; precision measures alert correctness. Set tradeoffs using consequences, review capacity and validation evidence.

Independent question

Missing a diseased plant is considered more costly than a false alarm, but staff can review only ten alerts daily. Which evaluation trade-off should be discussed?

Expected independent answer

Prioritize disease recall while checking precision and alert volume against review capacity. Choose thresholds using validation evidence and the stated consequences.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “One metric can be maximized without considering workload or other error types.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Compare validation precision/recall across thresholds, missed-positive consequences and review workload under realistic volume. More recall can overload review; fewer alerts can miss important cases. Choose a documented tradeoff before final testing rather than treating one metric as universally best.

Explanations for the additional workbook examples

W1 Apply the boundary

Which IDs are positive at .5 and .3?

Teacher explanation: At .5: A, B, C. At .3: A, B, C, D, E, including equality.

W2 Count the consequences

Give TP, FP, FN, TN at both thresholds.

Teacher explanation: At .5: 2, 1, 1, 2. At .3: 3, 2, 0, 1.

W3 Use the declared cost

Calculate costs with FP cost 1 and FN cost 4.

Teacher explanation: Costs are $1+4=5$ and $2+0=2$; .3 is preferred among these candidates.

W4 Apply independently

Scores/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare thresholds .5 and .2 using the same cost.

Teacher explanation: At .5 TP 1 FP 1 FN 1 TN 1 cost 5. At .2 TP 2 FP 1 FN 0 TN 1 cost 1. Equality is positive at both boundaries.

W5 Protect the final test

Why not select the threshold with the lowest final-test cost?

Teacher explanation: Selection uses the test answers, turning the test into development data and weakening its independent evaluation role.

Assess independent transfer

Selected case: Scores/references are .8/blocked,.5/clear,.2/blocked,.1/clear. Compare thresholds .5 and .2 using the same cost.

Selected case answer

Teacher explanation: At .5 TP 1 FP 1 FN 1 TN 1 cost 5. At .2 TP 2 FP 1 FN 0 TN 1 cost 1. Equality is positive at both boundaries.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Approve the Plant Classifier metric suite: accuracy for context, spotted precision/recall, confusion matrix, and subgroup slices.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 26: Reproducible Experiments. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use score and threshold in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: At .5 TP 1 FP 1 FN 1 TN 1 cost 5. At .2 TP 2 FP 1 FN 0 TN 1 cost 1. Equality is positive at both boundaries. Fresh retry .9/clear,.6/blocked,.4/clear,.2/blocked at threshold .4: TP 1 FP 2 FN 1 TN 0, cost 6 under FP 1 FN 4. Require equality and all counts. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Calibration as a group property. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/26, ai-4/28, ai-4/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.